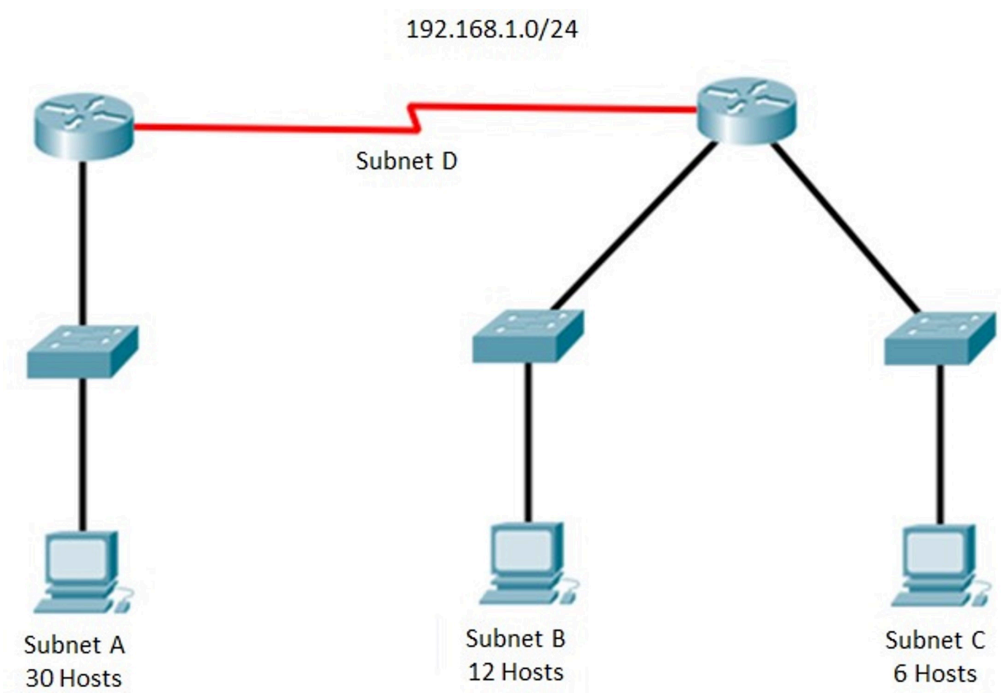
Review Test Submission: TSP 02u

User	Shazia Bano Patel
Course	Fall 2024 Introduction to Networks (NET-125-1001)
Test	TSP 02u
Started	9/15/24 8:37 PM
Submitted	9/15/24 9:15 PM
Status	Completed
Attempt Score	100 out of 100 points
Time Elapsed	38 minutes out of 2 hours
Results Displayed	Submitted Answers, Incorrectly Answered Questions

Question 1

5 out of 5 points



Refer to the exhibit. Use traditional subnetting to develop an addressing scheme based on the number of **hosts** using 192.168.1.0/24 as the starting point, then answer all of the questions below. Include the prefix length for the network

addresses only. Express the interval and octet using the format: 32, 4th

Determine the following: **Custom Address Map (CAM)**

Hint: Since the requirements are based on the number of hosts:

1. Use formula for Total Usable Addresses: $2^h - 2$
2. Set it greater than or equal to the largest network.
3. Solve for h.
4. The value of h represents the number of 'h' (host) bits in the CAM.

Selected Answer: N.N.N.ssshhhhh

Question 2 5 out of 5 points

Determine the following: **Interval and Transition Octet**

Record your answer using the format *interval, octet*. For example: 8, 2nd

Selected Answer: 32,4th

Question 3 4 out of 4 points

Determine the following: **Prefix Length**

Specify it as /NN, for example: /14

Selected Answer: /27

Question 4 4 out of 4 points

Determine the following: **Custom Subnet Mask**

Selected Answer: 255.255.255.224

Question 5 5 out of 5 points

Determine the following: **Wildcard Mask**

Selected Answer: 0.0.0.31

Question 6 4 out of 4 points

Determine the following: **Number of Subnets Possible**

Selected Answer: 8

Question 7 4 out of 4 points

Determine the following: **Total Addresses per Subnet**

Selected Answer: 32

Question 8 5 out of 5 points

Determine the following: **Total Usable Addresses per Subnet**

Selected Answer: 30

Question 9 4 out of 4 points

Subnet A: Network Address and Prefix Length

Hint 1: Start with the given address to subnet and apply the proper prefix.

Hint 2: Determine all network addresses first.

Hint 3: To determine the next network address:

- Add the interval to the transition octet of this network address.
- Note: The host bits of netwok addresses are always set to 0.

Record your answer using the following format as an example: 172.16.8.0/20

Selected Answer: 192.168.1.0/27

Question 10 4 out of 4 points

Subnet A: First Usable Address

Hint: Add 1 to octet 4 of the Network Address.

Do not include the prefix length.

Selected Answer: 192.168.1.1

Question 11 4 out of 4 points

Subnet A: Last Usable Address

Hint: Determine the Broadcast Address first, then subtract 1 from octet 4 of the broadcast.

Do not include the prefix length.

Selected Answer: 192.168.1.30

Question 12 4 out of 4 points

Subnet A: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 192.168.1.31

Question 13 4 out of 4 points

Subnet B: Network Address and Prefix Length

Hint 1: Add the interval to the transition octet of the previous network address and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 172.16.8.0/20

Selected Answer: 192.168.1.32/27

Question 14 4 out of 4 points

Subnet B: First Usable Address

Hint: Add 1 to octet 4 of the Network Address.

Do not include the prefix length.

Selected Answer: 192.168.1.33

Question 15 4 out of 4 points

Subnet B: Last Usable Address

Hint: Determine the Broadcast Address first, then subtract 1 from octet 4 of the broadcast.

Do not include the prefix length.

Selected Answer: 192.168.1.62

Question 16 4 out of 4 points

Subnet B: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 192.168.1.63

Question 17 4 out of 4 points

Subnet C: Network Address and Prefix Length

Hint 1: Add the interval to the transition octet of the previous network address and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 172.16.8.0/20

Selected Answer: 192.168.1.64/27

Question 18 4 out of 4 points

Subnet C: First Usable Address

Hint: Add 1 to octet 4 of the Network Address.

Do not include the prefix length.

Selected Answer: 192.168.1.65

Question 19 4 out of 4 points

Subnet C: Last Usable Address

Hint: Determine the Broadcast Address first, then subtract 1 from octet 4 of the broadcast.

Do not include the prefix length.

Selected Answer: 192.168.1.94

Question 20 4 out of 4 points

Subnet C: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 192.168.1.95

Question 21 4 out of 4 points

Subnet D: Network Address and Prefix Length

Hint 1: Add the interval to the transition octet of the previous network address and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 172.16.8.0/20

Selected Answer: 192.168.1.96/27

Question 22 4 out of 4 points

Subnet D: First Usable Address

Hint: Add 1 to octet 4 of the Network Address.

Do not include the prefix length.

Selected Answer: 192.168.1.97

Question 23 4 out of 4 points

Subnet D: Last Usable Address

Hint: Determine the Broadcast Address first, then subtract 1 from octet 4 of the broadcast.

Do not include the prefix length.

Selected Answer: 192.168.1.126

Question 24 4 out of 4 points

Subnet D: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 192.168.1.127

Sunday, September 15, 2024 9:15:55 PM EDT

← **OK**